



DATA ANALYSIS

7 Month Course Plan



Beginner to Job-Ready Data Analyst



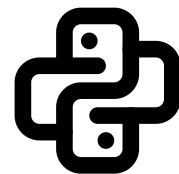
Excel



Statistics



Sql &
Database



Python

Data Analysis – Week-wise Course Plan (7 Months)

Total Duration: 28 Weeks

Level: Beginner to Job-Ready Data Analyst

FOUNDATION LEVEL (Weeks 1–8) Month 1:

Introduction to Data Analysis & Excel

Week 1: Introduction to Data Analysis

- What is Data Analysis?
- Types of data (structured & unstructured)
- Data analysis lifecycle
- Role & responsibilities of a Data Analyst
- Real-world business use cases
- Tools used in data analysis



Outcome: Clear understanding of the data analyst role

Week 2: Excel Basics for Data Analysis

- Excel interface & navigation

- Excel shortcuts & productivity tips
- Data types in Excel
- Importing data
- Basic formulas

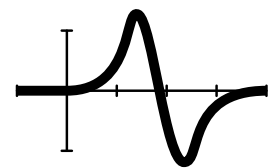
Outcome: Comfortable working with Excel



Week 3: Excel Data Cleaning & Functions

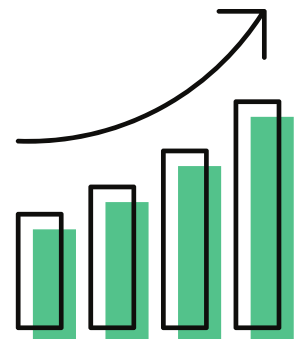
- Data cleaning techniques
- Handling missing & duplicate data
- Common Excel functions (SUM, IF, VLOOKUP/XLOOKUP)
- Text & date functions
- Data validation

Outcome: Ability to clean and prepare data



Week 4: Excel Analysis & Visualization

- Conditional formatting
- Pivot tables
- Pivot charts
- Basic data visualization
- Excel-based mini project



Outcome: Perform basic data analysis using Excel

Month 2: Statistics for Data Analysis

Week 5: Descriptive Statistics

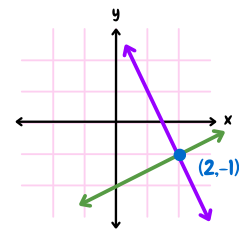
- What is statistics?
- Descriptive vs inferential statistics
- Mean, median, mode
- Range & percentiles
- Real-world examples



Outcome: Understand data summaries

Week 6: Probability & Data Distribution

- Probability basics
- Data distributions
- Variance & standard deviation
- Normal distribution
- Outliers & data spread



Outcome: Analyze data variability

Week 7: Correlation, Regression & Sampling

- Correlation concepts
- Linear regression basics
- Sampling techniques
- Bias & data accuracy
- Business examples



Outcome: Identify relationships in data

Week 8: Statistics Practical Applications

- Statistics using Excel
- Business problem-solving
- Case studies
- Foundation-level assessment



Outcome: Strong statistical foundation for analysis

ADVANCED LEVEL (Weeks 9–28)

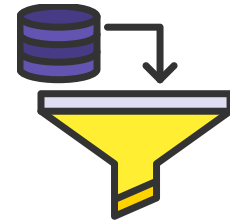
Month 3: SQL for Data Analysis

Week 9: Database Fundamentals & SQL Basics

- What is a database?
- Tables, rows & columns



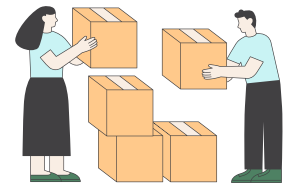
- SQL syntax basics
- SELECT & WHERE clauses
- Filtering data



Outcome: Query structured data

Week 10: SQL Aggregation & Grouping

- GROUP BY & HAVING
- Aggregate functions (SUM, AVG, COUNT)
- Sorting & limiting results
- Practical exercises



Outcome: Summarize large datasets

Week 11: SQL Joins & Subqueries

- Types of joins
- Subqueries
- Nested queries
- Real-world use cases



Outcome: Combine multiple datasets

Week 12: Advanced SQL Concepts

- Window functions
- Performance optimization basics
- SQL project
- Assessment

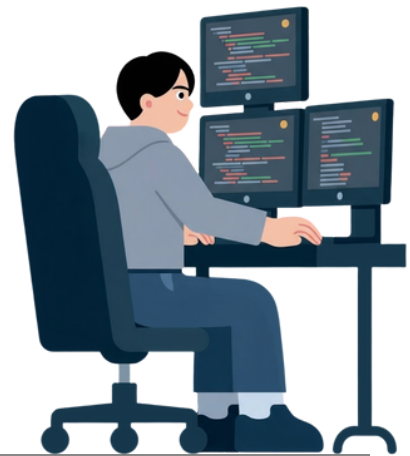
Outcome: Advanced SQL querying skills

Month 4: Python for Data Analysis

Week 13: Python Basics

- Introduction to Python
- Variables & data types
- Loops & conditions
- Functions
- Python for data analysis overview

Outcome: Python programming fundamentals



Week 14: NumPy & Pandas

- NumPy arrays
- Pandas Series & DataFrames

- Reading datasets
- Data selection & filtering

Outcome: Handle data using Python

Week 15: Data Cleaning & EDA

- Handling missing values
- Data transformation
- Exploratory Data Analysis (EDA)
- Descriptive statistics with Pandas



Outcome: Clean and explore datasets

Week 16: Data Visualization with Python

- Matplotlib basics
- Seaborn basics
- Charts & plots
- Real dataset visualization project

Outcome: Visualize insights using Python

Month 5: Data Visualization & BI Tools

Week 17: Visualization Principles

- Data visualization fundamentals
- Choosing the right chart
- Color & design principles
- Dashboard planning

Outcome: Design effective visualizations

Week 18: Power BI / Tableau Basics

- BI tool interface
- Data import
- Data modeling basics
- Simple visual creation



Outcome: Build basic BI dashboards

Week 19: Dashboards & KPI Reporting

- KPI creation
- Interactive dashboards
- Filters & slicers
- Business reporting

Outcome: Create professional dashboards

Week 20: Storytelling with Data

- Data storytelling concepts
- Business insights
- Report presentation
- Visualization project



Outcome: Communicate insights clearly

Month 6: Advanced Analytics & Data Ethics

Week 21: Hypothesis Testing

- Hypothesis concepts
- Null & alternative hypothesis
- Statistical testing basics
- Business examples



Outcome: Data-driven decision making

Week 22: A/B Testing & Time Series

- A/B testing concepts

- Metrics & analysis
- Time series data
- Trend & seasonality

Outcome: Analyze experiments & trends

Week 23: Forecasting & Business Analytics

- Forecasting basics
- Predictive analysis overview
- Business analytics use cases
- Case studies



Outcome: Apply analytics to business problems

Week 24: Data Management & Ethics

- Data quality management
- Data governance
- Data privacy & security
- GDPR basics
- Ethical data usage

Outcome: Responsible data handling



Month 7: Projects & Career Preparation

Week 25: Real-World Data Project – Phase 1

- Problem statement
- Data collection
- Data cleaning
- Analysis planning

Outcome: Project foundation

Week 26: Real-World Data Project – Phase 2

- Analysis execution
- Visualization
- Insight generation
- Final reporting



Outcome: Complete end-to-end project

Week 27: Case Studies & Portfolio

- Industry case studies
- Portfolio building

- GitHub / dashboard showcase
- Best practices

Outcome: Job-ready portfolio

Week 28: Career Preparation

- Resume building
- Interview questions (SQL, Python, Stats)
- Mock interviews
- Industry roadmap

Outcome: Job-ready Data Analyst



DATA ANALYSIS COURSE

Learn to Analyze & Visualize Data

Excel | SQL | Tableau

Data Cleaning | Data Exploration

Course Modules

Data Cleaning | Data Exploration | Statistical Analysis | Dashboards & Reports

ENROLL NOW!

CERTIFICATION PROVIDED | START YOUR CAREER IN DATA ANALYTICS!